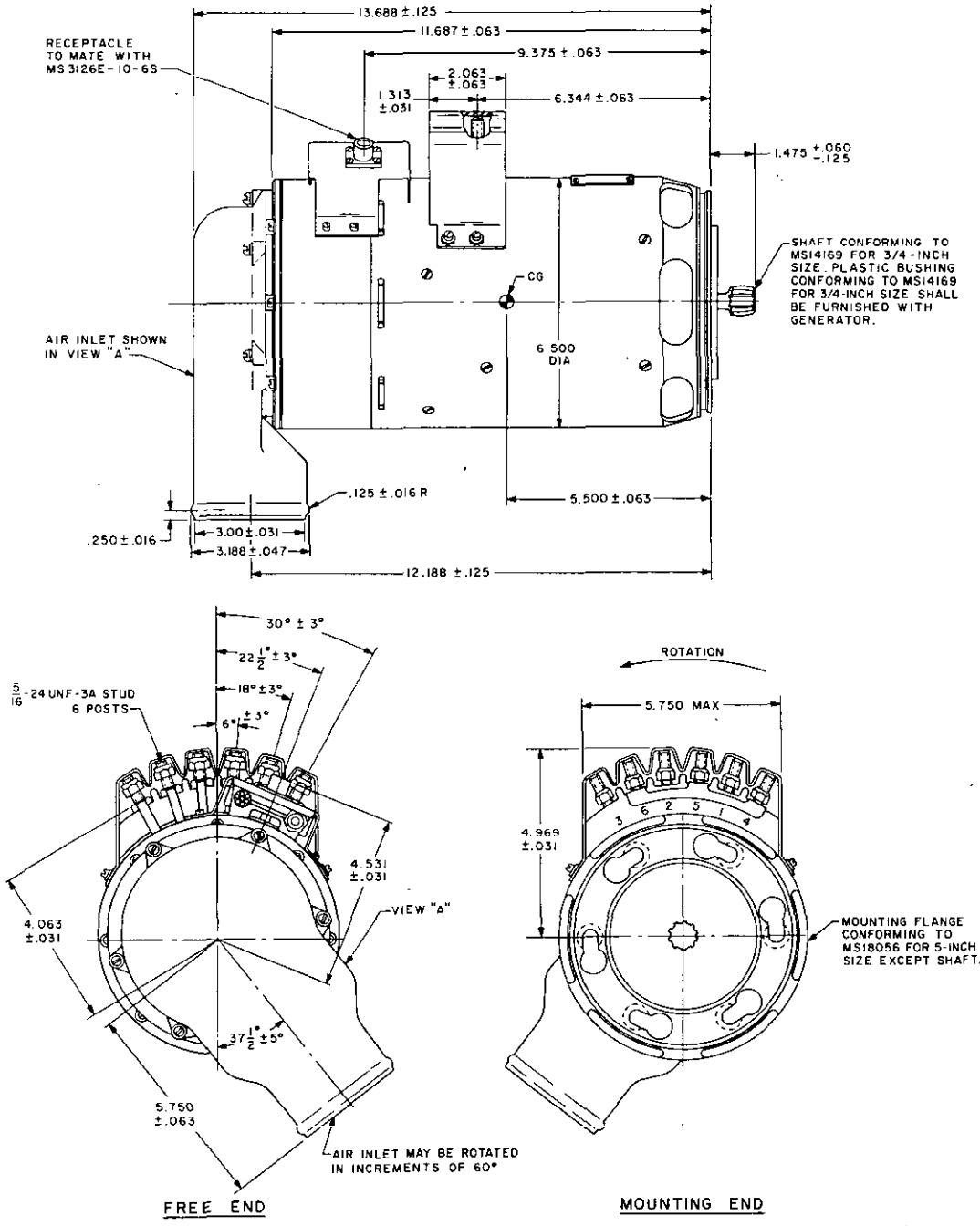
MILITARY SPECIFICATION SHEET

GENERATING SYSTEM, 20 KVA, 400 HERTZ,
ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS

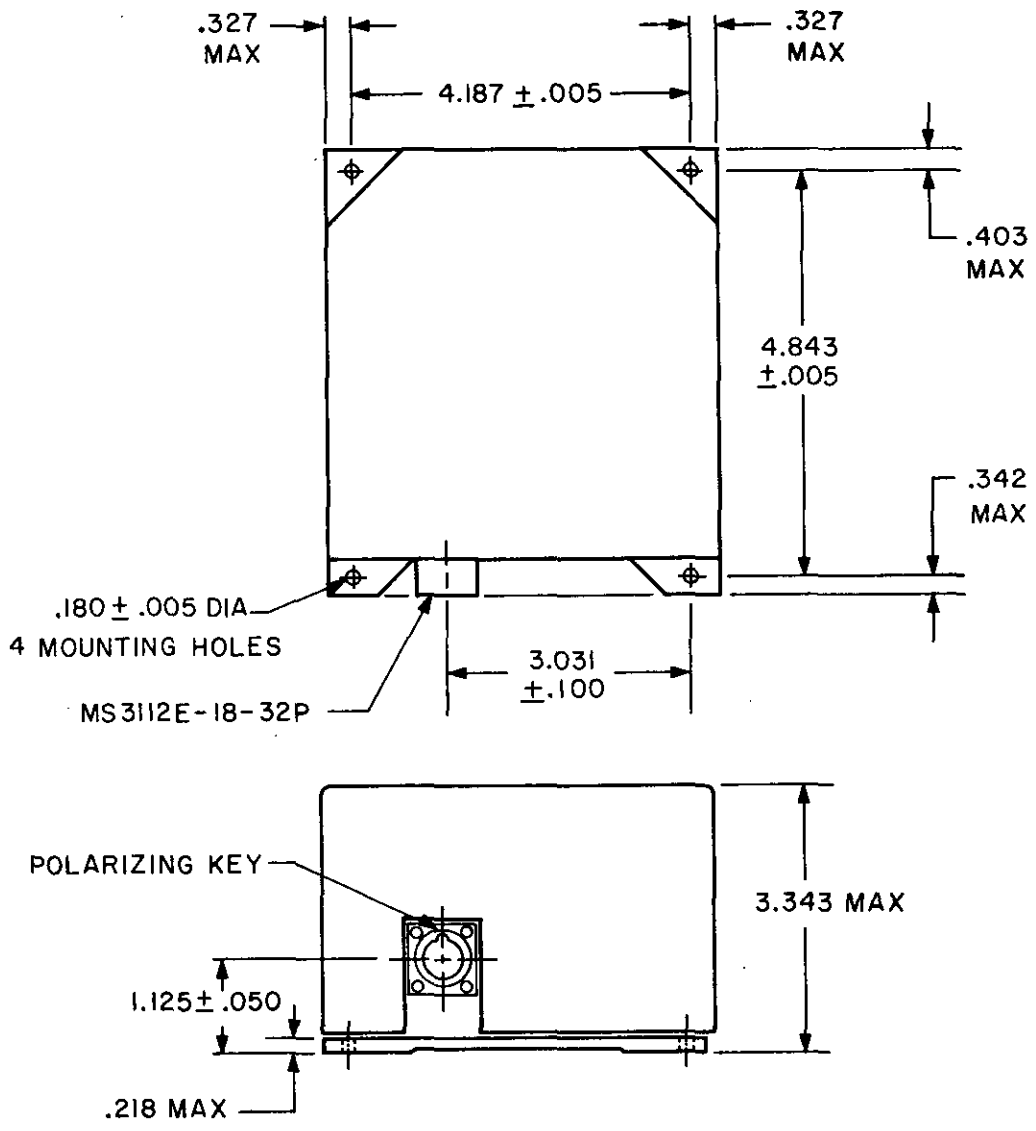
THE COMPLETE REQUIREMENTS FOR PROCURING THE GENERATOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-21480.



GENERATOR (M21480/17-1)

FSC 6115

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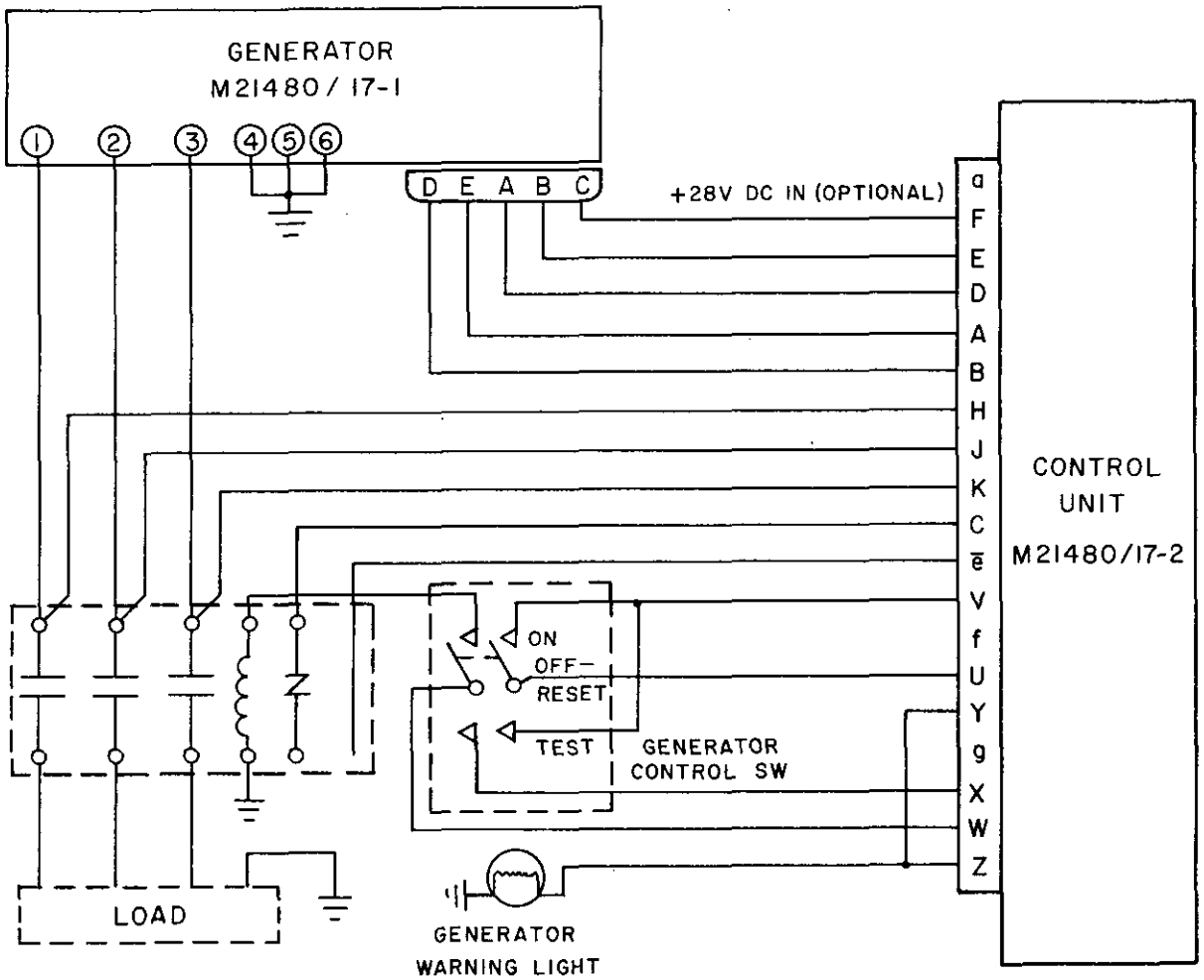


CONTROL UNIT, WT. 2.5 LBS. MAX.
(M21480/17-2)

IDENTIFICATION OF ITEMS	
ITEM	MILITARY PART NUMBER
System (consists of one Generator and one Control Unit)	M21480/17-3
Generator	M21480/17-1
Control Unit	M21480/17-2

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GENERATOR DATA	
RATED VOLTAGE	120/208
RATED OUTPUT (SEE PAGE 4)	20KVA (NOMINAL)
PHASES	3
SPEED RANGE	7600 - 8400 RPM
MAX. SPEED FOR REGULATION	10,000 RPM
OVERSPEED	11,000 RPM
RATED POWER FACTOR	0.75 TO 1.0
EFFICIENCY MIN. AT RATED LOAD	85%
MAX. WEIGHT	48.5 LBS
OVERHUNG MOMENT - MAX.	280 IN-LB
FREQUENCY	380 - 420 Hz
CLASS	C
SHEAR	1,000 - 1,400 IN-LB
FLEXIBLE DRIVE	REQUIRED
ROTOR MOMENT OF INERTIA	WR^2 - 0.3 LB FT^2



Test Circuit

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REQUIREMENTS:

1. All paragraphs of MIL-G-21480 shall apply except for the following deletions, additions, and substitutions:

3.4.8 System capacity (Rating). Delete and add: "The rating of the system shall be as shown on Figure 1 of the specification sheet. The nominal system rating shall be the same as the generator rating.

3.4.8.2 Overload capacity. Change "5 minutes" to "2 minutes".

3.4.9.3 Electric connections. Delete and add: "The Generator shall be supplied with connections as shown on the specification sheet."

3.4.9.6.1.1 Altitude rating chart. Change "10,000 hour life" to "1,000-hour life".

3.4.10 Control component design. Delete the first sentence.

3.4.10.5 Modular construction. Delete and substitute: "Modular construction shall be used."

3.4.10.6 Potting compounds. Delete.

3.4.13 Identification of product. Delete "assemblies and parts".

3.4.13.1 Nomenclature. Delete.

3.4.15 Provisions for maintainability. Delete and substitute: "During the duration of the contract, the contractor shall establish and conduct a maintainability program in accordance with MIL-STD-470 to the extent applicable to the state of hardware development on date of contract award."

3.4.16 Maintainability (Air Force). Delete this paragraph and all its subparagraphs.

3.4.17 Reliability. Delete and substitute: "The generating system shall have a minimum mean time between failures (MTBF) as follows:

(a) System - 3,000 hours
(b) Generator - 5,000 hours
(c) GCU - 7,800 hours"

3.5.1 Voltage limits. Delete "Figure 3" and substitute "Figure 2 of the specification sheet."

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3.5.4 Waveform. Delete and substitute: "The crest factor for each phase voltage waveform shall be 1.41 ± 10%. No single harmonic shall exceed 3% of the fundamental."

3.5.5 Integral control power. Delete "5 amps continuous capacity" and substitute "2 amperes continuous capacity".

3.5.7.2 Overvoltage. Delete "Figure 6" and substitute "Figure 2 of the specification sheet."

3.5.7.3 Undervoltage. Delete and substitute: "The undervoltage protective device shall disconnect the system from the load and deenergize the generator when any phase voltage drops to 90-105 volts for a time delay of 4 ± 1 seconds. The system shall be capable of being manually reset when all three phase voltages are above drop-out level and before each phase exceeds 105 volts."

3.5.7.4 Underfrequency. Delete and substitute: "The underfrequency protective device shall function to disconnect the system from the load in less than one second. The frequency drop-out and automatic pick-up shall occur between the values of 350-360 Hz, with a minimum of 3 Hz between drop-out and pick-up."

3.5.7.5 Overfrequency. Delete.

3.5.7.6 Feeder fault. Delete and substitute: "The G.C.U. shall include provision for feeder-fault protection in accordance with MIL-G-21480/16."

3.5.8 Electromagnetic interference. Delete and substitute: "Conducted and radiated radio interference shall be in accordance with MIL-I-6181 with the following exceptions:

(a) Conducted radio interference excursions up to 6,000 microvolts over the range of 0.15 to 0.25 Mhz.
(b) Radiated radio interference excursions up to 102 db above 1 microvolt per Mhz over the range of 0.25 to 2.0 Mhz."

4.5.3.2 Acceptance tests. Delete and add: "Each G.C.U. shall be tested for voltage regulation at 400 Hz, no load, 1.0, 1.5 and 2.0 P.U. loads. Limits: Voltage - 115 ± 2.5 volts; voltage modulation 0 to 1.0 P.U. load shall be 0.75% maximum."

4.5.6 Exciter polarity reversal test. Delete.

Table I - Test Methods. Delete requirement for Exciter Polarity Reversal Test (4.5.6) under Generator Acceptance Tests.

4.5.13 Waveform. The limits of 3.5.4 shall apply.

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4.5.23.1.1 Generator (Vibration). The amplitude of vibration monitored at the free end of the generator shall be limited to 25 g's. The resonant frequency shall be greater than 190 Hz as determined by Procedure VII of MIL-E-5272.

4.5.23.1.2 Control components (Vibration). Delete requirements above 500 Hz.

4.5.23.2 Acceptance test. Delete.

4.5.28 Maximum pressure drop. Change "400 ± 15 inches" to "200 ± 15 inches".

4.5.32 Vast compatibility demonstration. Delete.

4.5.33 Reliability and maintainability testing (Air Force). Delete.

4.5.33.1 Delete.

4.5.33.2 Delete.

NOTES:

1. Dimensions in inches. Tolerances: ± .03 inches unless otherwise specified.

2. All outline dimensions are limiting dimensions only.

Preparing Activity:

NAVY - AS

Project No. 6115-N454

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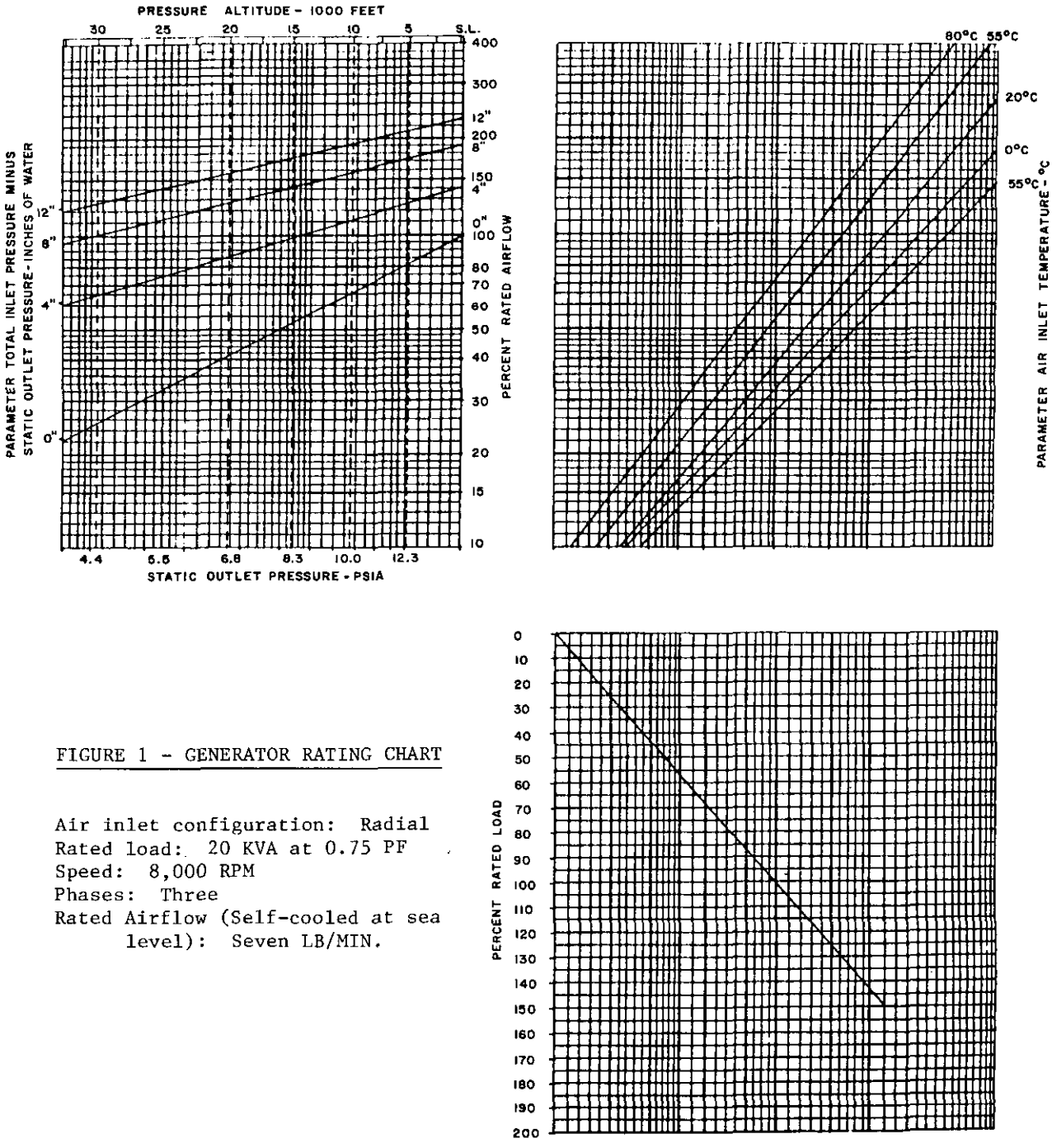


FIGURE 1 - GENERATOR RATING CHART

Air inlet configuration: Radial
Rated load: 20 KVA at 0.75 PF
Speed: 8,000 RPM
Phases: Three
Rated Airflow (Self-cooled at sea level): Seven LB/MIN.

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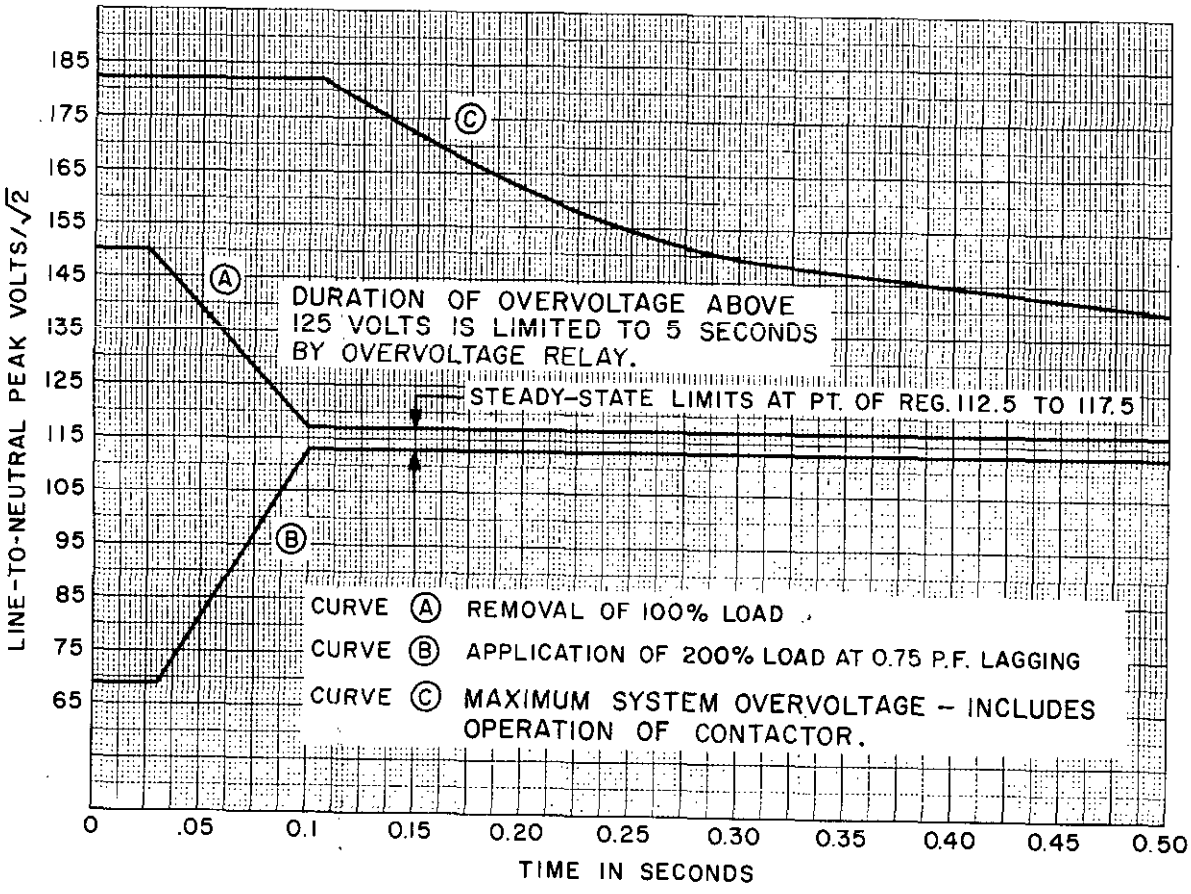


FIGURE 2 - Limits of Potential

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